Technology Stack

Date	1 July 2026
Team ID	XXXXXX
Project Name	Emotion Detection and Learning Support Engine
Maximum Marks	2 Marks

Define Your Technology Stack

The **Emotion Detection and Learning Support Engine (EDLSE)** is developed using a modern, scalable technology stack consisting of frontend technologies, backend frameworks, artificial intelligence and machine learning libraries, databases, deployment platforms, and external tools. These technologies were selected for their performance, accuracy, scalability, maintainability, and ease of development.

Technology Stack Details

S. No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	**Frontend / Client-Side**	**HTML5, CSS3, Streamlit**	Provides a responsive, interactive, and user-friendly interface for emotion detection, learning recommendations, and dashboard visualization.
2	**Backend / Server-Side**	**Python**	Handles application logic, emotion detection, AI model integration, recommendation generation, and report creation efficiently.
3	**AI/ML Framework**	**TensorFlow, Keras, Scikit-learn**	Develops and deploys deep learning models for emotion recognition and personalized learning support.

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4	**Computer Vision**	**OpenCV, MediaPipe**	Captures and processes facial expressions from webcam or uploaded images for facial emotion recognition.
5	**Natural Language & Speech Processing**	**NLTK, TextBlob, Transformers, Librosa**	Performs sentiment analysis, text preprocessing, speech feature extraction, and voice emotion recognition.
6	**Database / Data Storage**	**SQLite / JSON Files / Local Storage**	Stores user profiles, emotion history, learning progress, reports, and application data in a lightweight and efficient format.
7	**Cloud / Hosting / Deployment**	**Streamlit Community Cloud**	Enables free cloud deployment, easy updates, and remote accessibility for users.
8	**Version Control & CI/CD**	**Git & GitHub**	Supports source code management, version control, collaboration, and project tracking.
9	**Reporting & Visualization**	**Matplotlib, Plotly, FPDF (or ReportLab)**	Generates emotion analytics, performance charts, and downloadable PDF reports.
10	**Third-Party Libraries / Other Tools**	**NumPy, Pandas, Pillow**	Supports data manipulation, image processing, numerical computations, and overall application performance.